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Course Code: A8851

**VARDHAMAN**

COLLEGE OF ENGINEERING

III B.Tech I Semester Continuous Assessment - II, October - 2025

(Regulations: VCE-R22)

Data Science for Engineers**(Common to CSE & IT)**

Date: 24 October 2025

Time: 2 Hours

Maximum Marks: 30

Answer all Questions in Part-A

Answer any FOUR Questions in Part-B

Course Outcomes with Bloom's Levels:

CO#	CO Statement	Bloom's Level (L#)
CO1	Identify the various requirements for data science process	L1
CO2	Choose an appropriate database required for processing data	L4
CO3	Demonstrate the data science methodology and text mining approaches	L2
CO4	Make use of data science tools to visualize the insights of data	L3
CO5	Apply various data visualization techniques using Tableau over Google Sheets	L3

Questions:

Questions.

PART-A (Multiple Choice / Fill in the Blanks / Match the Following / Short Answer Type Questions)					
			CO#	L#	Marks
1.	a)	Discuss the ETL Process.	CO3	L2	1M
	b)	List any three tools used for model building.	CO3	L2	1M
	c)	Explain the term Text Mining.	CO3	L1	1M
	d)	How does NumPy assist in handling numerical computations in data science?	CO4	L1	1M
	e)	Discuss the use of sample () in python.	CO4	L1	1M
	f)	Which function in python is used to delete the missing values from the dataset?	CO4	L2	1M
	g)	What is the purpose of joins and blends in Tableau?	CO5	L2	1M
	h)	Explain custom table calculations in Tableau and how they differ from basic calculations.	CO5	L2	1M
	i)	What are the advantages of using Tableau for data visualization and analysis?	CO5	L1	1M
	j)	What is the concept of data blending in Tableau, and how is it used?	CO5	L2	1M

PART-B (Descriptive Questions)				
2.	Write a short note on phases of Data Analytics life cycle.	CO3	L3	5M
3.	Elaborate the below Text Analysis subtasks. (i)Part-of-Speech Tagging (ii)Stemming (iii)Lemmatization	CO3	L3	5M
4.	Describe the implementation of DBSCAN clustering in Python, highlighting its process and key steps.	CO4	L4	5M
5.	Explain Principal Component Analysis (PCA) in detail.	CO4	L3	5M
6.	How are dashboards designed in Tableau, and what principles should be followed to create effective visualizations?	CO5	L4	5M
7.	What are the special chart types supported by Tableau, and how can a waterfall chart be created in Tableau? Explain the steps involved.	CO5	L3	5M



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SCHEMA OF VALUATION

Name of the Subject	: Data Science for Engineers	Subject Code	: A8851
Programme	: B.Tech	Year	: III
Examination	: Regular / Supplementary. CAT-2	Date of Examination	: 24/10/2025
Max. Marks	: 30	Time of Examination	: FN

① ② ETL process are i) Extraction

ii) Transformation

iii) Loading

process Name 0.5 m

Explanation 0.5 m

1 mark

① ③ Tools used for model building are SAS, Apache Spark, RIGAL

(Any 3 Tools — 1 m)

① ④ Text mining refers to the process of extraction of meaningful information, patterns and insights from unstructured text data using techniques from natural language processing (NLP), machine learning & statistics.

(Definition — 1 m)

① ⑤ Numpy provides i) Efficient Array operation
ii) Mathematical Functions

(Any 2 Advantages — 1 mark)

①② Sample() :- This is a built-in function displays a random rows from the dataset. Passing any integer in the function will display that many random rows
(Use — 1 mark)

①⑧ dropna() :- This function is used to delete the missing values from the data set. (function name — 1 mark)

①⑧ Purpose of join :- A join is a way to combine two tables based on a common field (key).

Purpose of Blends :- A blend is used to combine data from two different data sources.

Purpose of join — 0.5 m } 1 mark
" " Blend — 0.5 m }

①⑨ Custom table calculation in Tableau are performed on aggregated data in visualization, while basic calculations operate at the row level in the underlying data set before aggregation. (Difference 1 mark)

①⑩ Advantages of Tableau

i) Tableau creates an automatic dashboard

ii) Tableau can handle millions & billions of data without slowing down the dashboard performance.

(Any 2 advantages 1 mark)

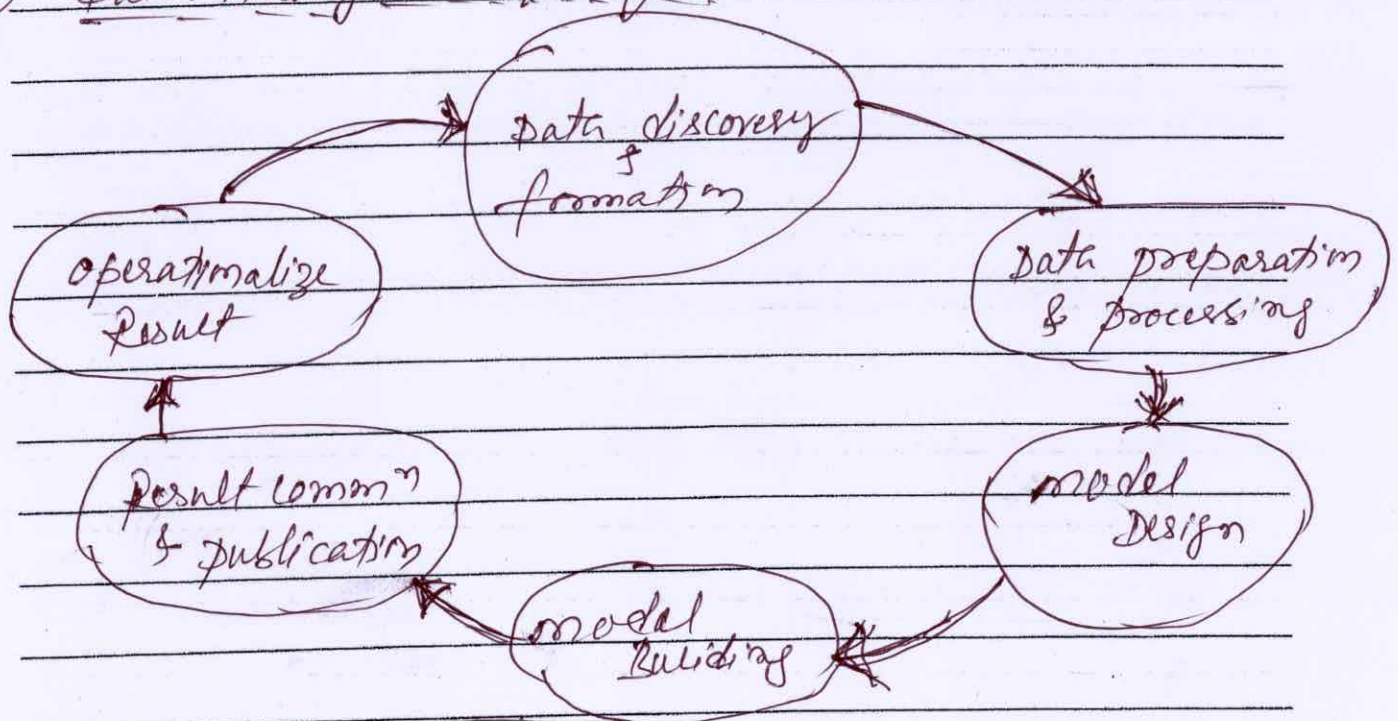


①(i) Data blending in Tableau combines data from multiple sources by linking them on a common field.

It is used by designating a primary data source and linking it with secondary data sources on a common field, enabling the creation of a unified analysis without merging the dataset physically.

Data blending 0.5 m
Use 0.5 m
1 mark

② Data Analytics life cycle.



Phases 2 marks
Explanation 3 marks



③ Text Analysis Subtasks

i) part of speech Tagging:- It can be known as the process of assigning the parts of speech to the selected words

Ex:-

noun	verb	preposition	noun
Rama	went	to	school

ii) Stemming:- It strips the word and extract the base form of that particular word. This step is done by removing the affixes of the word

Ex:- stem of the word sleeping and sleep's is sleep

iii) Lemmatization:- It finds the root word of the given word. The output of this technique is called "lemma"

Ex:- plays: play.

part of speech Tagging	2 marks	} 5 marks
Stemming	1.5 marks	
Lemmatization	1.5 marks	

④ DBSCAN clustering Algorithm implementation in python

Input: from sklearn.cluster import DBSCAN
from sklearn.datasets import make_blobs

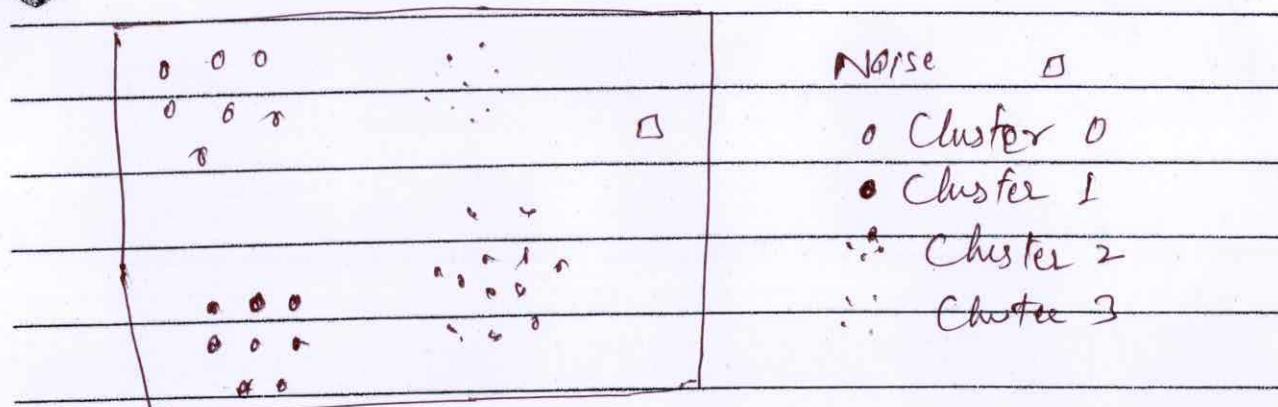
X, y = make_blobs (random_state = 9, n_samples = 25)

dbscan = DBSCAN (eps = 2, min_samples = 2)

cls = dbscan.fit_predict(X)

print ("cluster membership values: \n {}".format(cls))

plot_labeled_scatter (X, cls + 1, ['noise', 'cluster0', 'cluster1', 'cluster2'])



Process & Key Stages

- i) Core samples which are apart by a distance of ϵ unit are put into one cluster
- ii) The data points not making it into any cluster are said to be as noise.
- iii) Boundary points are the points that fall within the ϵ unit distance but cannot be termed as core points.

Implementation 2 marks
Process & Key steps 3 marks } 5 marks

(5) Principal Component Analysis (PCA)

- i) It is a technique which extracts a new set of variables from an existing wide range of variables
- ii) These variables are regarded as the principal components
- iii) The linear collection of the initial variables is known as principal component.
- iv) The ~~first~~ principal component describes the maximum variance in the dataset
- v) The second principal component seeks to outline the remaining variance in the data set.

vi) The next principal component aims to describe the variation that has remained unexplained by the first two principal components and so on.

Use	2 marks
Explanation with diagram	4 marks
5 marks	

6) Dashboard Design and principles

A dashboard is an interactive representation of limited worksheets and all the essential information. Dashboards are mainly used to compare various data at one go.

Steps to create dashboard.

- 1) Go to the new dashboard worksheet by navigating the sample superstore sheet.
- 2) Drag the sheets from the left of that dashboard section to the canvas area.
- 3) Check show dashboard title and add the dashboard title.
- 4) Fit the view with respect to the available space.
- 5) Place the profit legend with respect to the sheet.
- 6) Take time to interact with the dashboard.

Dashboard design principles

- i) Be the audience.
- ii) Relevant dashboard
- iii) Use minimal type of color
- iv) layout design.

Design steps	3.0 marks
principle	2.0 marks
5 marks	



⑦ Special charts supported by Tableau.

- ① Motion chart
- ② Rump chart
- ③ Waterfall chart.

Waterfall chart :-

It depicts the flow between certain statistics

It is used to analyse a measure its rise and fall over the years

Steps:- i) Drag the order date and profit to columns and rows shelves respectively.

Change the format of the order date to month/year and for the profit, convert the quick table calculation to running total.

ii) Under the marks pane, change the mark type to Gantt Bar

iii) Create a calculated field named negative profit and drag it to size under the marks pane. This field will fill the Gantt chart.

iv) Drag profit to the type color under marks pane to have different color.

Special chart types	1 mark
Waterfall chart	1 mark
Steps	3 marks
	5 marks



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